

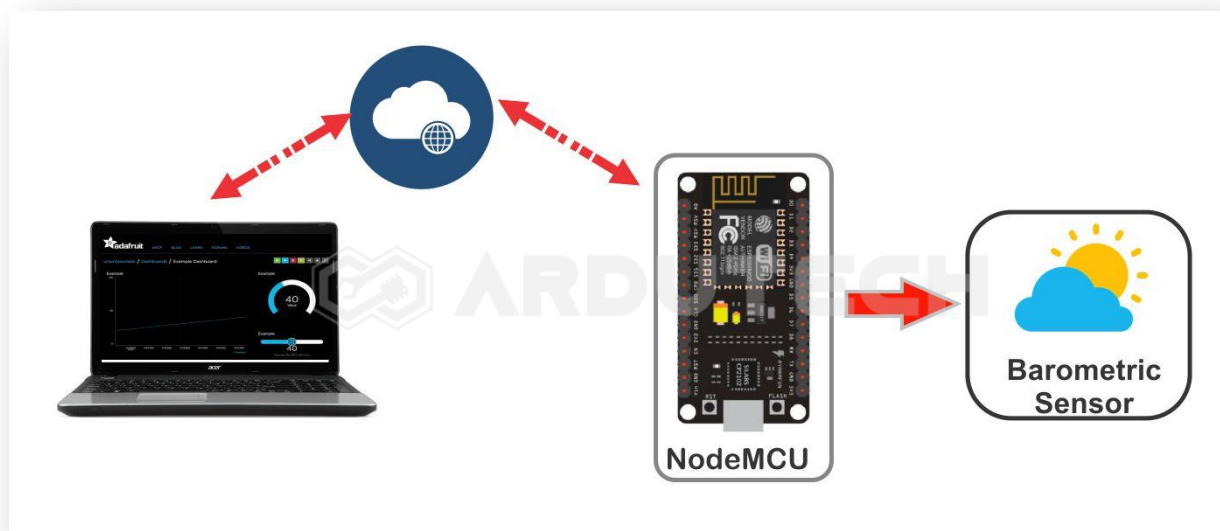
## Stasiun Cuaca dg Adafruit IO dan BME280

### Deskripsi

Membuat proyek IoT (*Internet of Things*) untuk membaca data sensor barometric BME280 kemudian data dikirim secara kontinyu ke Adafruit IO dengan protokol MQTT. Data berupa nilai temperatur, kelembaban, tekanan udara dan altitude (ketinggian).

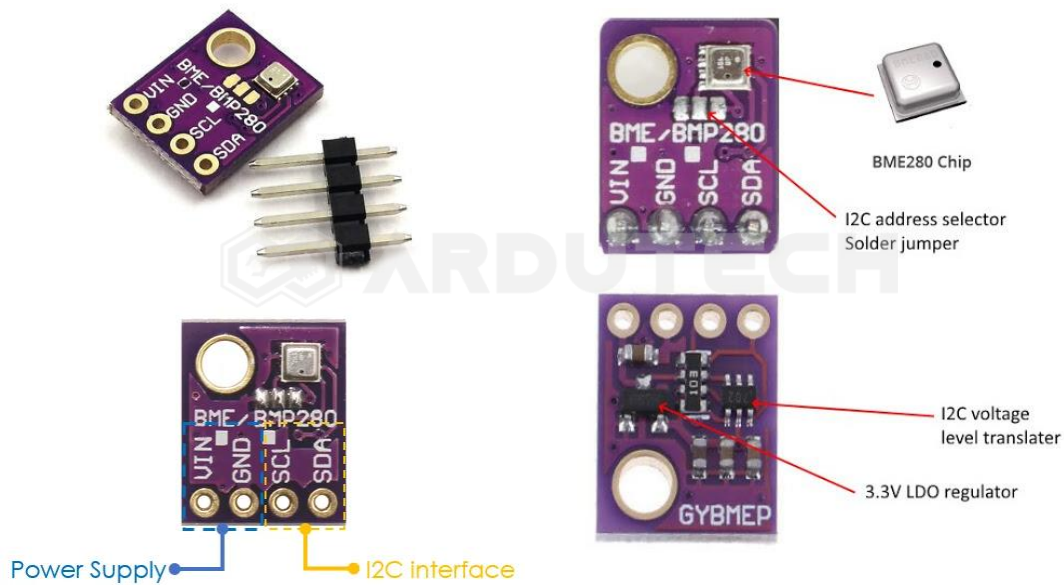
### Cara Kerja

Sensor BME280 mengukur besaran temperature, kelembaban, tekanan udara dan altitude (ketinggian) kemudian mengirimkan data hasil pengukuran ke NodeMCU ESP8266. NodeMCU terhubung dengan jaringan internet (WiFi) kemudian mengirimkan datanya ke Adafruit IO dan ditampilkan hasil pengukurannya.



BME280 merupakan sensor berukuran fisik kecil dengan kemampuan mengukur tekanan barometrik (berat udara), temperature, kelembaban dan juga ketinggian (altitude). Sensor ini dapat berkomunikasi dengan model I2C maupun SPI & pada proyek ini nanti kita akan memakai komunikasi I2C yang memanfaatkan kaki SDA dan SCL.

Sensor BMP280 sudah dikemas dalam sebuah modul yang terdiri dari 4 pin seperti terlihat pada gambar berikut ini.



### Spesifikasi :

- Altimeter : Mengukur ketinggian tempat dari permukaan air laut (0 – 9,2KM) dg akurasi - +1M
- Thermometer : Mengukur suhu udara/ruangan ( Range : -40 sampai +85 C )
- Humidity : mengukur kelembaban udara 0 – 100 %
- Atmospheric Pressure Meter : Mengukur tekanan udara disuatu tempat (Range : 300 hPa – 1100 hPa dg akurasi 12Pa)
- Komunikasi interface I2C dan SPI
- Address I2C : 0x76
- Akurasi relatif : 0.12hPa / 1 meter
- Akurasi absolut : 1hPa

Terdapat 4 pin koneksi pada sensor BME280 dengan keterangan sebagai berikut :

- VIN : tegangan input (3.3 – 5 V)
- GND : Ground
- SDA : pin data (komunikasi I2C)
- SDL : pin clock (komunikasi I2C)

Adafruit IO adalah sistem yang disediakan oleh Adafruit (pembuat modul – modul sensor, perangkat elektronik, robot dll) untuk membuat sebuah proyek/aplikasi IoT (*Internet of Things*) . Adafruit IO ini boleh dipakai oleh siapa saja karena sifatnya free.



### Kebutuhan Bahan

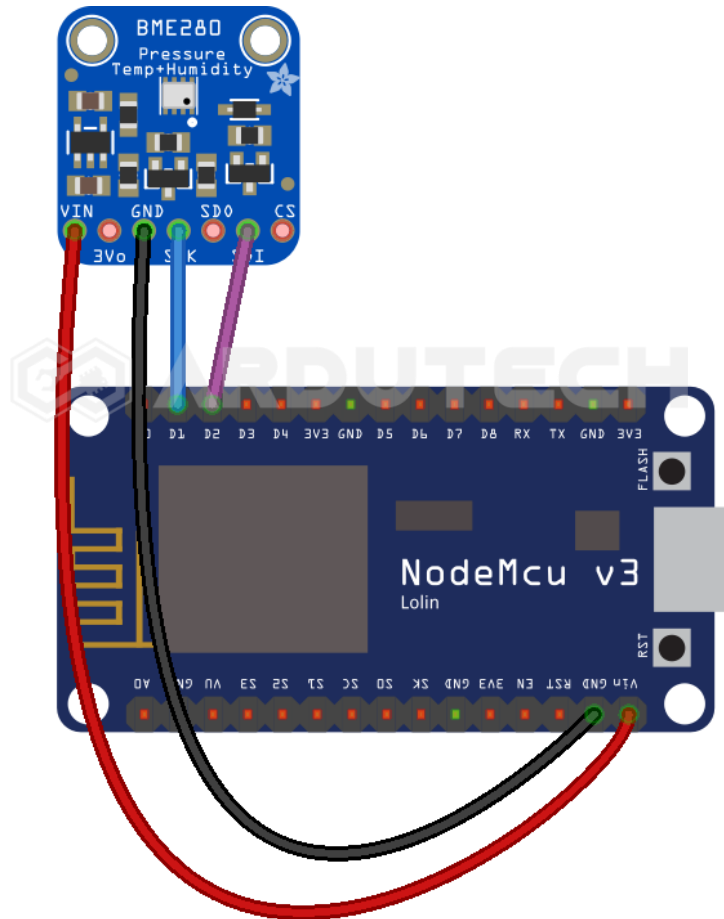
- NodeMCU V3
- Sensor Barometric Pressure BME280
- Kabel konektor
- Kabel micro USB



### Kebutuhan Software

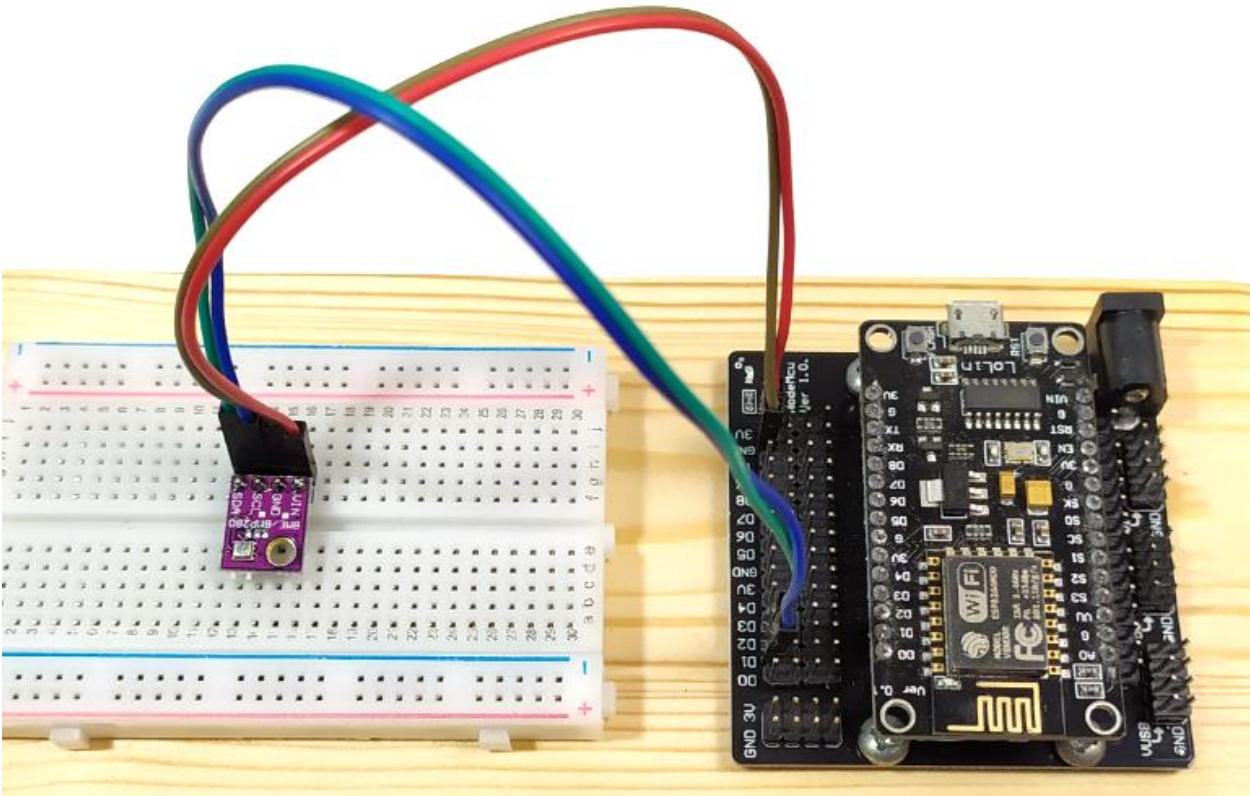
- Arduino IDE

### Rangkaian/Skematik



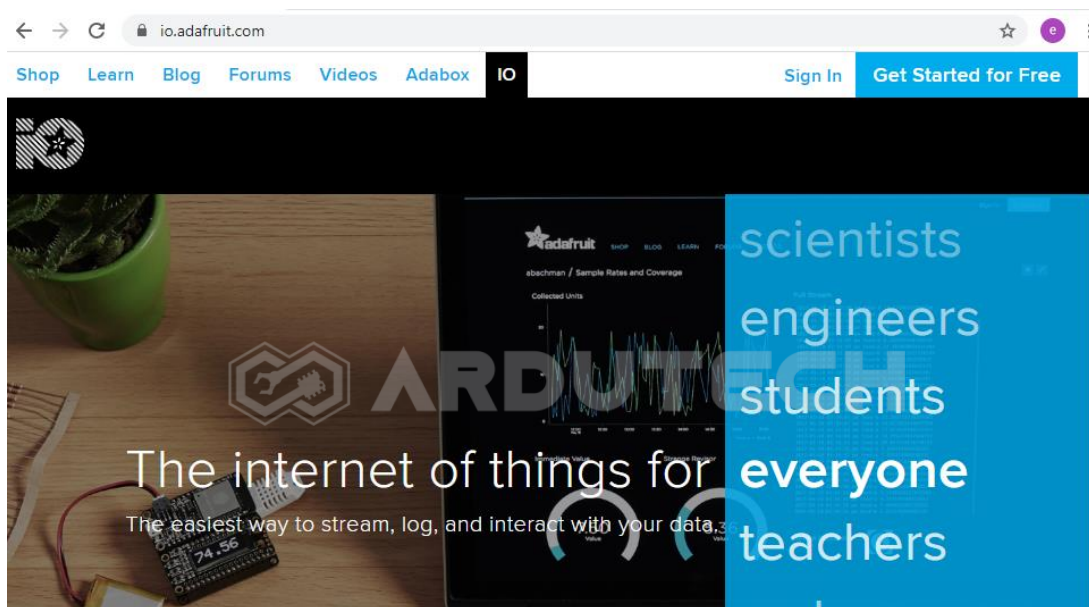
Koneksi NodeMCU dengan Sensor BME280 :

| NodeMCU | BME280 |
|---------|--------|
| D1      | SCL    |
| D2      | SDA    |
| 3V3     | VCC    |
| GND     | GND    |



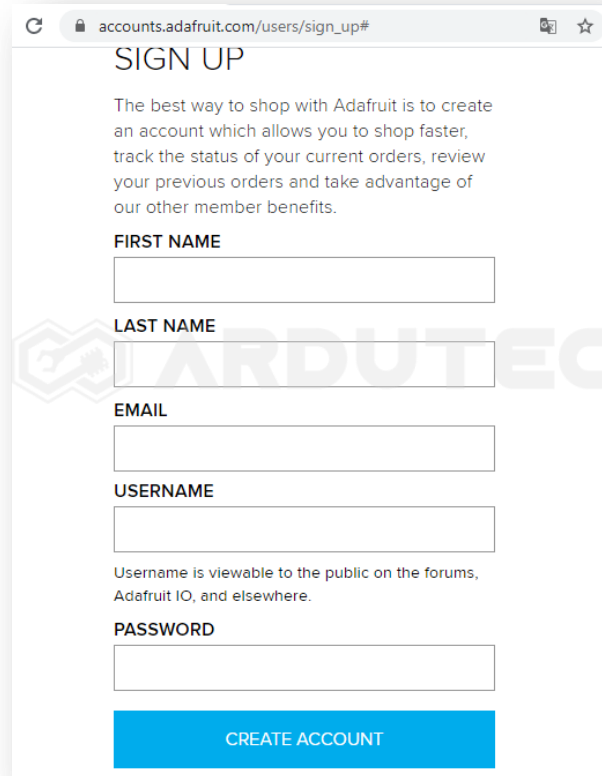
## Membuat Antarmuka di Adafruit

Silakan masuk ke halaman <https://io.adafruit.com/>



Jika belum mempunyai akun silakan membuat akun terlebih dahulu. Siapkan sebuah akun email aktif.

Klik pada **“Get Started for free”** di bagian pojok kanan atas.



**SIGN UP**

The best way to shop with Adafruit is to create an account which allows you to shop faster, track the status of your current orders, review your previous orders and take advantage of our other member benefits.

**FIRST NAME**

**LAST NAME**

**EMAIL**

**USERNAME**

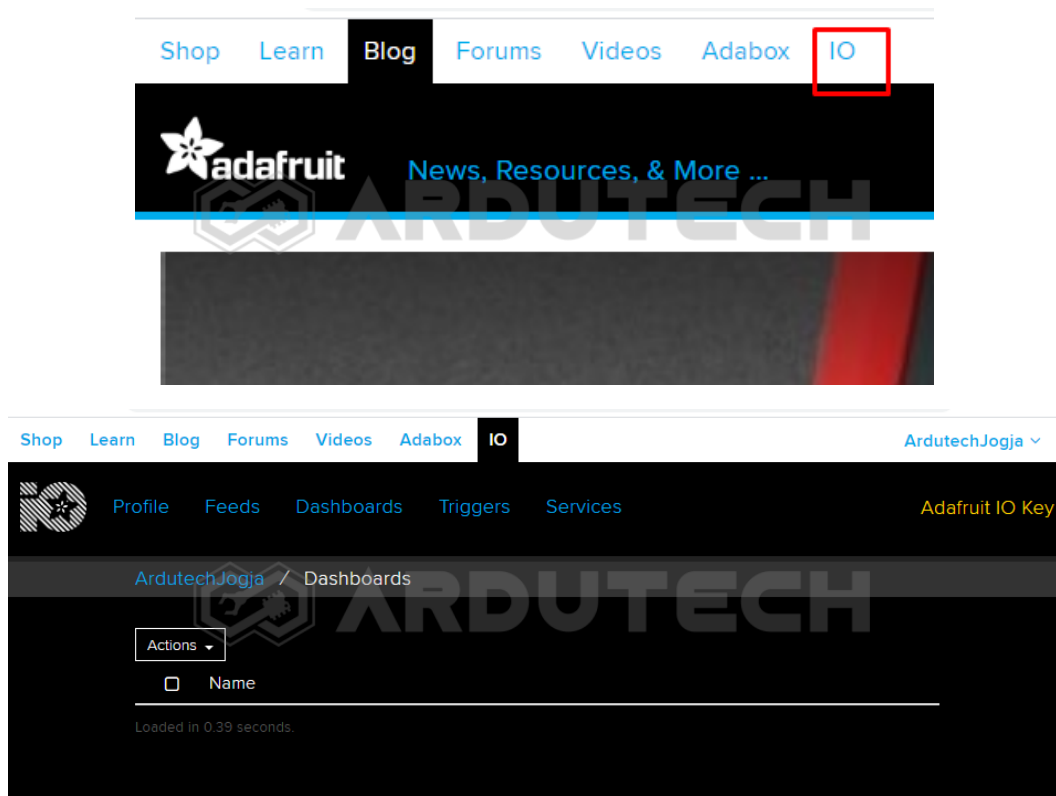
Username is viewable to the public on the forums, Adafruit IO, and elsewhere.

**PASSWORD**

**CREATE ACCOUNT**

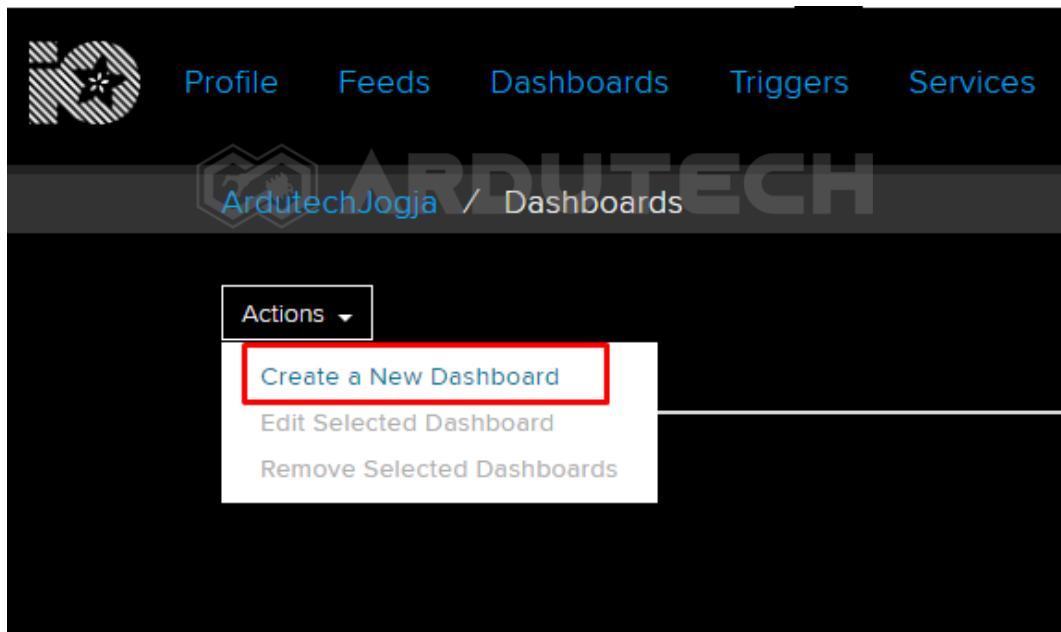
Isi data – data yang diperlukan kemudian klik **“CREATE ACCOUNT”** dan selanjutnya ikuti petunjuknya.

Masuk ke menu **IO**.



Klik pada tombol **“Actions”** kemudian pilih **“Create a New Dashboard”**

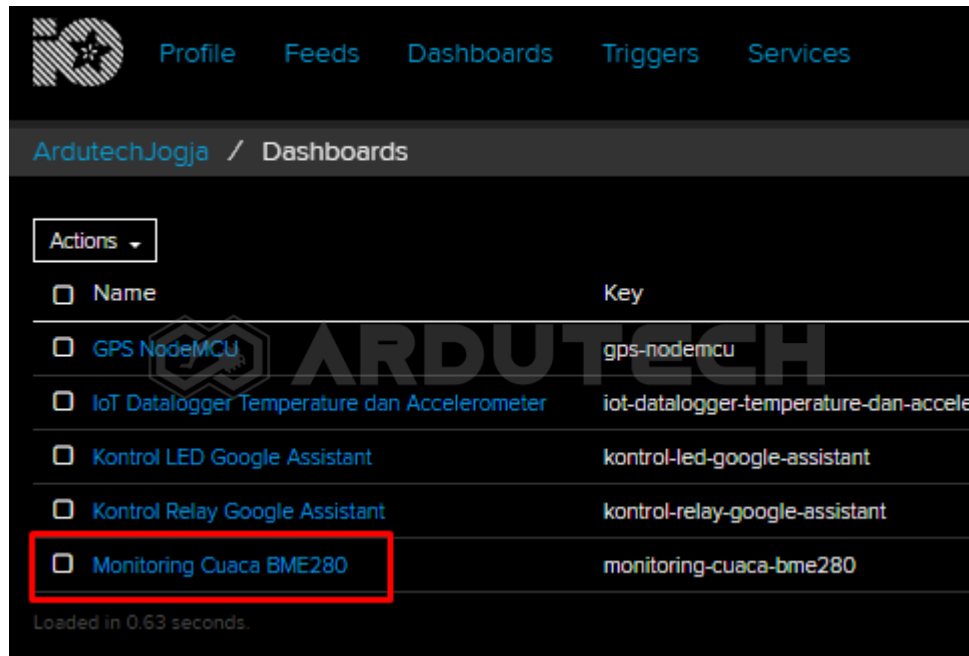




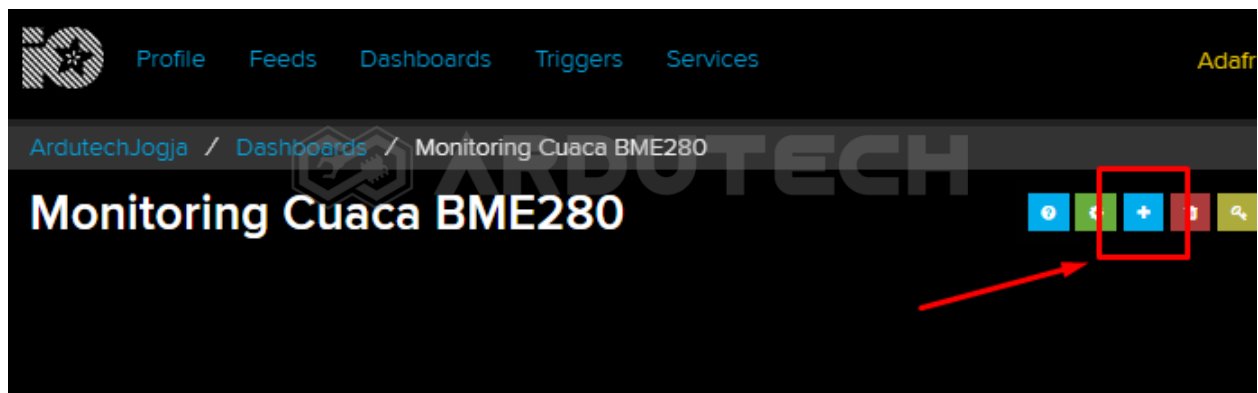
Selanjutnya akan muncul jendela untuk memberi nama, silakan beri nama pada dashboard yang akan dibuat, misalnya **“Monitoring Cuaca BME280”** dan klik **“Create”**.

A screenshot of a modal window titled 'Create a new Dashboard'. It features a close button (X) in the top right corner. The form contains two input fields: 'Name' and 'Description'. The 'Name' field is filled with the text 'Monitoring Cuaca BME280'. The 'Description' field is currently empty. At the bottom right of the modal, there are two buttons: 'Cancel' and 'Create'. The 'Create' button is highlighted in blue.

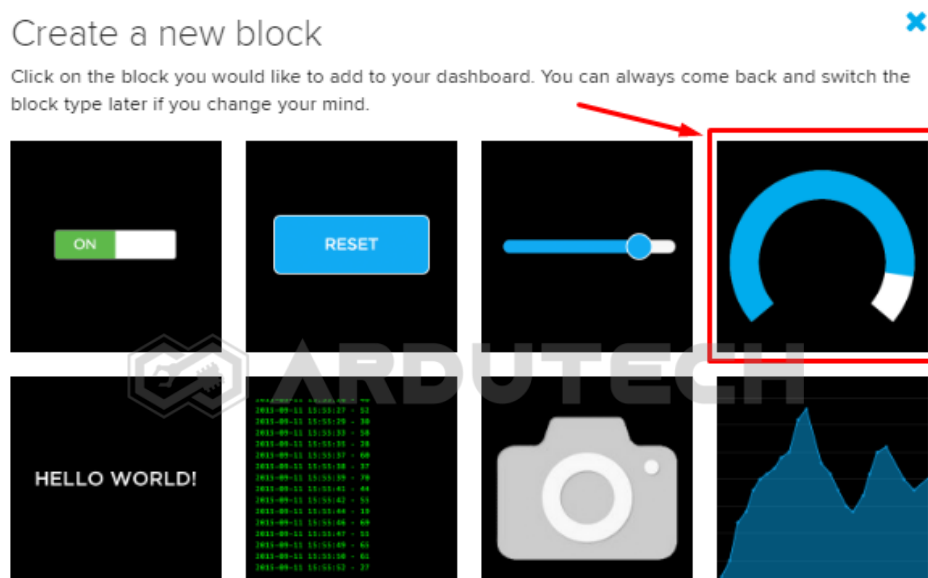
Setelah klik tombol **“Create”** , akan muncul dashboard baru dengan nama **“Monitoring Cuaca BME280”**.



Klik “Monitoring Cuaca BME280” kemudian pilih tombol “+” untuk membuat blok baru.



Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).





Selanjutnya muncul tampilan untuk memilih '**Chose feed**'. Ini semacam "Topik" dalam komunikasi MQTT. Jadi supaya dikenali oleh 'subscriber' (penerima perintah) maka 'publisher' harus menentukan 'topik' nya. Silakan tulis "**BMETemp**" kemudian klik tombol "**Create**".

## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

|                      |                                      |                                       |
|----------------------|--------------------------------------|---------------------------------------|
| <input type="text"/> | <input type="text" value="BMETemp"/> | <input type="button" value="Create"/> |
| Group / Feed         | Last value                           | Recorded                              |

Muncul feed BMETemp.

## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

|                                             |                                                  |                                       |
|---------------------------------------------|--------------------------------------------------|---------------------------------------|
| <input type="text"/>                        | <input type="text" value="Enter new feed name"/> | <input type="button" value="Create"/> |
| Group / Feed                                | Last value                                       | Recorded                              |
| <input checked="" type="checkbox"/> BMETemp |                                                  | 1 minute                              |
| <input type="checkbox"/> GPS                |                                                  | 4 days                                |
| <input type="checkbox"/> gpslat             | 0.000000                                         | 4 days                                |
| <input type="checkbox"/> gpslng             | 0.000000                                         | 4 days                                |
| <input type="checkbox"/> GSMloc             |                                                  | 4 days                                |

Klik/pilih **BMETemp**. Selanjutnya klik "**Next Step >**" sehingga muncul tampilan **Block settings** :

## Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Temperature

Gauge Min Value

0

Gauge Max Value

100

Gauge Width

50px

Gauge Label

Celsius

Low Warning Value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

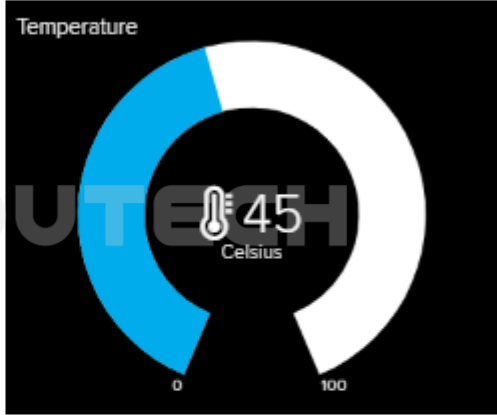
Decimal Places

2

Number of decimal places to display when value is a number. Defaults to 2.

Block Preview

Temperature



Gauge

A gauge is a read only block type that shows a fixed range of values.

Test Value

45

Isi pada kolom yang tersedia :

- Block Title : Temperature (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Gauge Label : Celsius

Jika ingin menambahkan icon bisa dengan mengaktifkan "Show icon" dibagian bawah.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

Number of decimal places to display when value is a number. Defaults to 2.

☒ Show Icon  
When checked, show an icon with the value.

Icon

Show this icon next to the value.

[Previous step](#) [Create block](#)

Klik tombol “**Create block**”. Selesai pembuatan block untuk temperature.



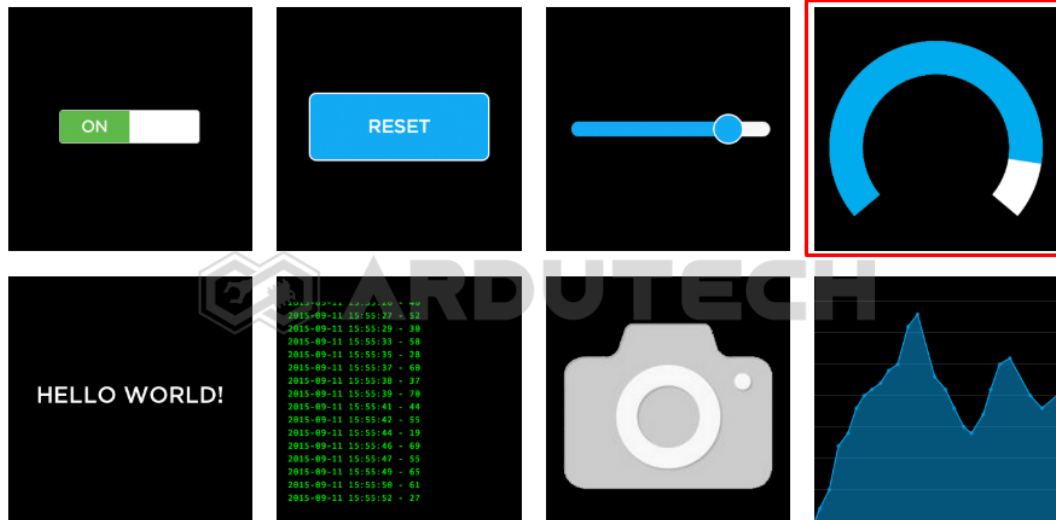
Kita akan membuat 3 block dengan tampilan yang sama yaitu “Gauge”. Ulangi langkah – langkah sebelumnya sehingga nanti ada 4 buah Gauge.

Klik tombol plus “+” lagi untuk menambahkan sebuah Gauge lagi ke dashboard.

Setelah tombol “+” di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).

## Create a new block

Click on the block you would like to add to your dashboard. You can always come back and switch the block type later if you change your mind.



Selanjutnya muncul tampilan untuk memilih 'Chose feed'. Ini semacam "Topik" dalam komunikasi MQTT. Silakan tulis "MBEHumi" kemudian klik tombol "Create".

## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Feed name input:

| Group / Feed | Last value | Recorded |
|--------------|------------|----------|
|--------------|------------|----------|

Muncul feed MBEHumi.

## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Search field:

Enter new feed name:

| Group / Feed                                | Last value | Recorded        |
|---------------------------------------------|------------|-----------------|
| <input checked="" type="checkbox"/> MBEHumi |            | less than a ... |
| <input type="checkbox"/> MBETemp            |            | 3 minutes       |
| <input type="checkbox"/> GPS                |            | 4 days          |
| <input type="checkbox"/> gpslat             | 0.000000   | 4 days          |
| <input type="checkbox"/> gpslng             | 0.000000   | 4 days          |
| <input type="checkbox"/> GSMloc             |            | 4 days          |

Klik/pilih **BMEHumi**. Selanjutnya klik “**Next Step** > ” sehingga muncul tampilan **Block settings** :

✕

## Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

**Block Title (optional)**

**Gauge Min Value**

**Gauge Max Value**

**Gauge Width**

**Gauge Label**

**Low Warning Value**

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

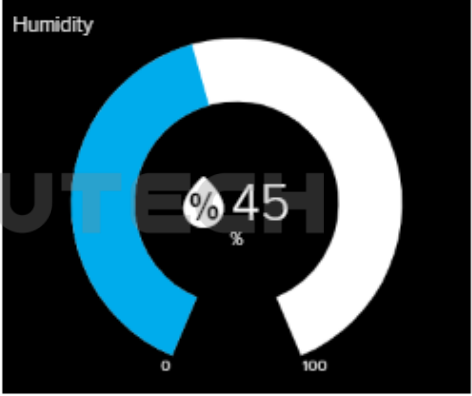
**High Warning Value**

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

**Decimal Places**

Number of decimal places to display when value is a number. Defaults to 2

**Block Preview**



**Gauge** A gauge is a read only block type that shows a fixed range of values.

**Test Value**

Isi pada kolom yang tersedia :

- Block Title : Humidity (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 100
- Gauge Width : 50px
- Gauge Label : %

Jika ingin menambahkan icon bisa dengan mengaktifkan “Show icon” dibagian bawah.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

Number of decimal places to display when value is a number. Defaults to 2.

☒ Show Icon  
When checked, show an icon with the value.

Icon

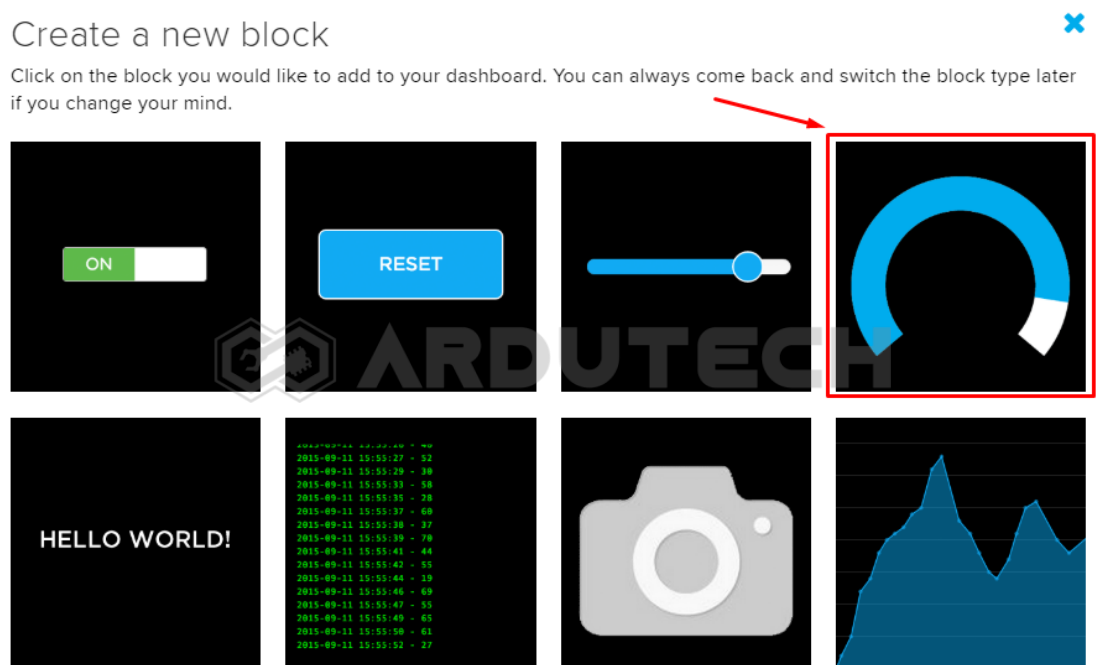
Show this icon next to the value.

[< Previous step](#) [Create block](#)

Klik tombol **“Create block”**. Selesai pembuatan block untuk humidity.

Klik tombol plus **“+”** lagi untuk menambahkan sebuah Gauge lagi ke dashboard.

Setelah tombol **“+”** di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).



Selanjutnya muncul tampilan untuk memilih **‘Chose feed’**. Ini semacam **“Topik”** dalam komunikasi MQTT. Silakan tulis **“BMEPress”** kemudian klik tombol **“Create”**.



## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

| Group / Feed | Last value | Recorded |
|--------------|------------|----------|
|--------------|------------|----------|

Muncul feed BMEPress.

## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

| Group / Feed                                 | Last value | Recorded        |
|----------------------------------------------|------------|-----------------|
| <input type="checkbox"/> BMEHumi             | 🔒          | 5 minutes       |
| <input checked="" type="checkbox"/> BMEPress | 🔒          | less than a ... |
| <input type="checkbox"/> BMETemp             | 🔒          | 8 minutes       |
| <input type="checkbox"/> GPS                 | 🔒          | 4 days          |
| <input type="checkbox"/> gpslat              | 🔒 0.000000 | 4 days          |
| <input type="checkbox"/> gpslng              | 🔒 0.000000 | 4 days          |
| <input type="checkbox"/> GSMloc              | 🔒          | 4 days          |

Previous step

Next step >

Klik/pilih **BMEPress**. Selanjutnya klik “**Next Step >**” sehingga muncul tampilan **Block settings** :

## Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Pressure

Gauge Min Value

300

Gauge Max Value

1200

Gauge Width

50px

Gauge Label

hPa

Block Preview

Pressure



Low Warning Value

Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

0

Gauge A gauge is a read only block type that shows a fixed range of values.

Test Value

45

Isi pada kolom yang tersedia :

- Block Title : Pressure (opsional)
- Gauge Min Value : 300
- Gauge Max Value : 1200
- Gauge Width : 50px
- Gauge Label : hPa

Jika ingin menambahkan icon bisa dengan mengaktifkan "Show icon" dibagian bawah.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

Number of decimal places to display when value is a number. Defaults to 2.

☒ Show Icon  
When checked, show an icon with the value.

Icon

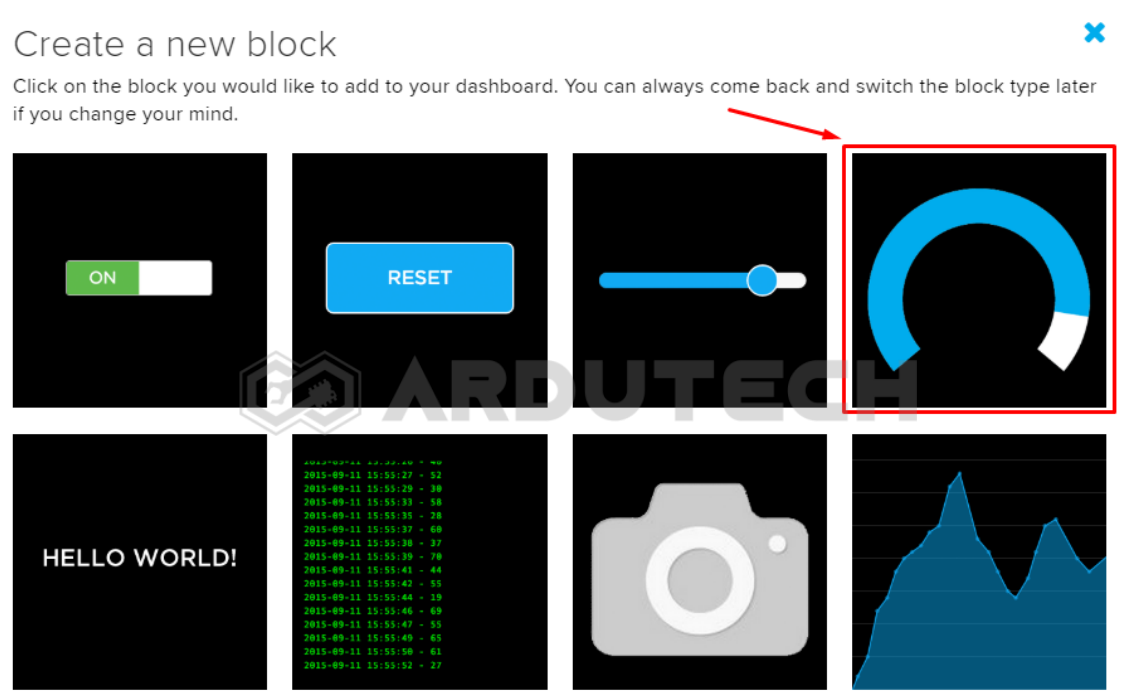
Show this icon next to the value.

[Previous step](#) [Create block](#)

Klik tombol **“Create block”**. Selesai pembuatan block untuk pressure.

Klik tombol plus **“+”** lagi untuk menambahkan sebuah Gauge lagi ke dashboard.

Setelah tombol **“+”** di-klik akan muncul tampilan bermacam ‘widget’. Ada tombol (Toggle), tampilan Gauge, Slider, Text dll. Pilih **Gauge** (pojok kanan atas).



Selanjutnya muncul tampilan untuk memilih **‘Chose feed’**. Ini semacam **“Topik”** dalam komunikasi MQTT. Silakan tulis **“BMEAlti”** kemudian klik tombol **“Create”**.

## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

| Group / Feed | Last value | Recorded |
|--------------|------------|----------|
|--------------|------------|----------|

Muncul feed BMEAlti.

## Choose feed

**Gauge:** A gauge is a read only block type that shows a fixed range of values.

If you have lot of feeds, you may want to use the search field. You can also create a feed quickly below.

Create

| Group / Feed                                | Last value | Recorded        |
|---------------------------------------------|------------|-----------------|
| <input checked="" type="checkbox"/> BMEAlti |            | less than a ... |
| <input type="checkbox"/> BMEHumi            |            | 9 minutes       |
| <input type="checkbox"/> BMEPress           |            | 5 minutes       |
| <input type="checkbox"/> BMETemp            |            | 12 minutes      |
| <input type="checkbox"/> GPS                |            | 4 days          |
| <input type="checkbox"/> gpslat             | 0.000000   | 4 days          |
| <input type="checkbox"/> gpslng             | 0.000000   | 4 days          |
| <input type="checkbox"/> GSMloc             |            | 4 days          |

Previous step

Next step >

Klik/pilih **BMEAlti**. Selanjutnya klik **"Next Step >"** sehingga muncul tampilan **Block settings** :

## Block settings

In this final step, you can give your block a title and see a preview of how it will look. Customize the look and feel of your block with the remaining settings. When you are ready, click the "Create Block" button to send it to your dashboard.

Block Title (optional)

Low Warning Value

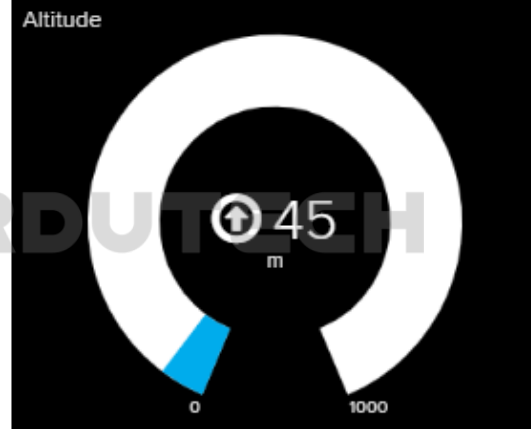
Optional. If no low warning value is given, the gauge will only change color when the value is out of bounds.

High Warning Value

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

Block Preview



Gauge A gauge is a read only block type that shows a fixed range of values.

Test Value

Isi pada kolom yang tersedia :

- Block Title : Altitude (opsional)
- Gauge Min Value : 0
- Gauge Max Value : 1000
- Gauge Width : 50px
- Gauge Label : m

Jika ingin menambahkan icon bisa dengan mengaktifkan "Show icon" dibagian bawah.

High Warning Value

45

Optional. If no high warning value is given, the gauge will only change color when the value is out of bounds.

Decimal Places

2

Number of decimal places to display when value is a number. Defaults to 2.

☒ Show Icon

When checked, show an icon with the value.

Icon

arrow-circle-o-up

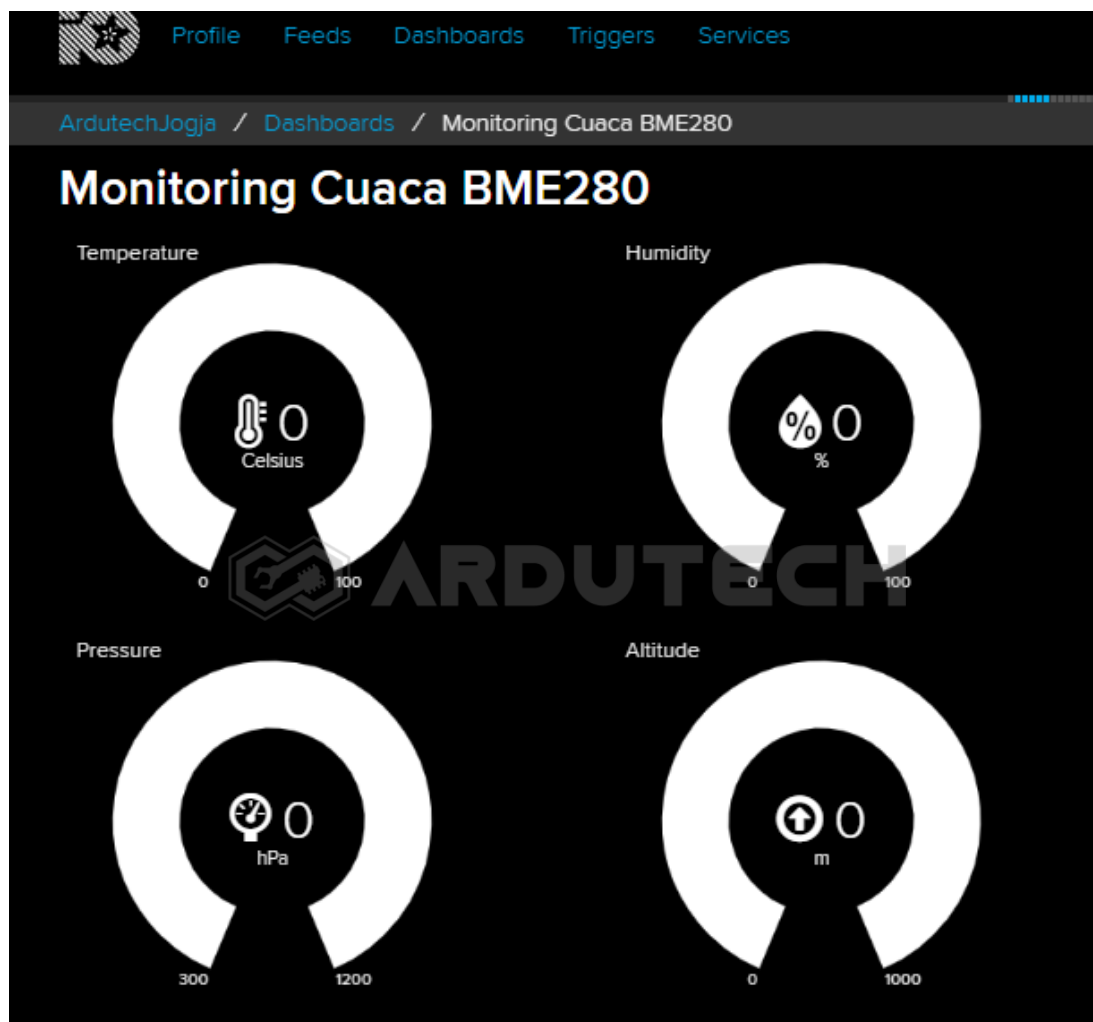
Show this icon next to the value.

&lt; Previous step

Create block

Klik tombol “**Create block**”. Selesai pembuatan block untuk altitude.

Ok sudah selesai pembuatan 4 buah block untuk menampilkan temperature, kelembaban, pressure dan altitude.



Selanjutnya kita perlu kode (token) untuk mengisi variabel username dan active key di program Arduino yang akan dibuat. Klik bagian “**Adafruit IO Key**” di pojok kanan atas.

## YOUR ADAFRUIT IO KEY

Your Adafruit IO Key should be kept in a safe place and treated with the same care as your Adafruit username and password. People who have access to your Adafruit IO Key can view all of your data, create new feeds for your account, and manipulate your active feeds.

If you need to regenerate a new Adafruit IO Key, all of your existing programs and scripts will need to be manually changed to the new key.



Username

ArdutechJogja

Active Key

ruu\_wUXJ670DJNEIIBQWERTY567MNB

REGENERATE KEY

[Hide Code Samples](#)

Arduino

```
#define IO_USERNAME "ArdutechJogja"
#define IO_KEY      "ruu_wUXJ670DJNEIIBQWERTY567MNB"
```

Perhatikan untuk **Username** dan **Active Key** ini penting, karena nanti dipakai ketika pemrograman dengan Arduino IDE. Catat/simpan kode tersebut.

Nah untuk pembuatan antarmuka pada Adafruit sudah selesai, sekarang kita siapkan program Arduino IDE-nya.

## Program/Source Code dengan Arduino IDE

Program pada proyek ini memerlukan library :

- ESP8266WebServer.h
- Adafruit\_MQTT.h
- Adafruit\_MQTT\_Client.h
- Adafruit\_Sensor.h
- Adafruit\_BME280.h
- Wire.h

Buka/jalankan Arduino IDE kemudian buat lembar kerja baru. Tulis kode program berikut.

```
/* *****
 * Project Stasiun Cuaca dg Adafruit dan BME280
 * Board   : NodeMCU ESP8266 V3
 * Input    : BME280
 * Output   : Adafruit IO
 * 99 Proyek IoT
 * www.ardutech.com
 * ***** /
#include <ESP8266WebServer.h>
```

```
#include <Wire.h>
#include <Adafruit_Sensor.h>
#include <Adafruit_BME280.h>
#include "Adafruit_MQTT.h"
#include "Adafruit_MQTT_Client.h"
#define SEALEVELPRESSURE_HPA (1013.25)
Adafruit_BME280 bme;
float temp, humi, pres, alti;

//---GANTI SESUAI DENGAN JARINGAN WIFI
//---HOTSPOT ANDA
#define WLAN_SSID      "ArdutechWiFi"  // Nama Hotspot/WiFi
#define WLAN_PASS      "12345678"      // Password
#define AIO_SERVER      "io.adafruit.com"
#define AIO_SERVERPORT  1883

//---GANTI SESUAI DENGAN USER NAME ADAFRUIT ANDA
#define AIO_USERNAME    "ArdutechJogja"

//---GANTI SESUAI DENGAN IO KEY ADAFRUIT ANDA
#define AIO_KEY          "aio_mAXJ90DJnEIBVj0IDmG6BIMZqMAW"

//*****
WiFiClient client;

Adafruit_MQTT_Client  mqtt(&client,      AIO_SERVER,      AIO_SERVERPORT,
AIO_USERNAME, AIO_KEY);

Adafruit_MQTT_Publish temperature = Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/BMETemp");

Adafruit_MQTT_Publish humidity    = Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/BMEHumi");

Adafruit_MQTT_Publish pressure    = Adafruit_MQTT_Publish(&mqtt,
AIO_USERNAME "/feeds/BMEPress");

Adafruit_MQTT_Publish altit = Adafruit_MQTT_Publish(&mqtt, AIO_USERNAME
"/feeds/BMEAlti");

//=====

void MQTT_connect() {
```



```

int8_t ret;

if (mqtt.connected()) {
    return;
}

Serial.print("Connecting to MQTT... ");

uint8_t retries = 3;

while ((ret = mqtt.connect()) != 0) { // connect will return 0 for
connected
    Serial.println(mqtt.connectErrorString(ret));
    Serial.println("Retrying MQTT connection in 5 seconds...");
    mqtt.disconnect();
    delay(5000); // wait 5 seconds
    retries--;
    if (retries == 0) {
        // basically die and wait for WDT to reset me
        while (1);
    }
}

Serial.println("MQTT Connected!");
}

//=====================================================
void setup() {
    Serial.begin(9600);
    delay(100);
    bme.begin(0x76);
    Serial.println(); Serial.println();
    Serial.print("Connecting to ");
    Serial.println(WLAN_SSID);
    WiFi.begin(WLAN_SSID, WLAN_PASS);
    while (WiFi.status() != WL_CONNECTED) {
        delay(500);
        Serial.print(".");
    }
}

```

```
}  
Serial.println();  
Serial.println("WiFi connected");  
}  
//======  
void loop() {  
  MQTT_connect();  
  temp = bme.readTemperature();  
  humi = bme.readHumidity();  
  pres = bme.readPressure() / 100.0F;  
  alti = bme.readAltitude(SEALEVELPRESSURE_HPA);  
  if (! temperature.publish(temp)) {  
    Serial.println(F("Publish Temperature Failed"));  
  } else {  
    Serial.println(F("Publish Temperature OK!"));  
  }  
  if (! humidity.publish(humi)) {  
    Serial.println(F("Publish humidity Failed"));  
  } else {  
    Serial.println(F("Publish Humidity OK!"));  
  }  
  if (! pressure.publish(pres)) {  
    Serial.println(F("Publish Pressure Failed"));  
  } else {  
    Serial.println(F("Publish Pressure OK!"));  
  }  
  if (! altit.publish(alti)) {  
    Serial.println(F("Publish Altitude Failed"));  
  } else {  
    Serial.println(F("Publish Altitude OK!"));  
  }  
  Serial.print("Temperature=");
```

```
Serial.println(temp);  
Serial.print("humidity=");  
Serial.println(humi);  
Serial.print("Pressure=");  
Serial.println(pres);  
Serial.print("Altitude=");  
Serial.println(alti);  
Serial.println();  
delay(30000);  
}
```

## PERHATIKAN !

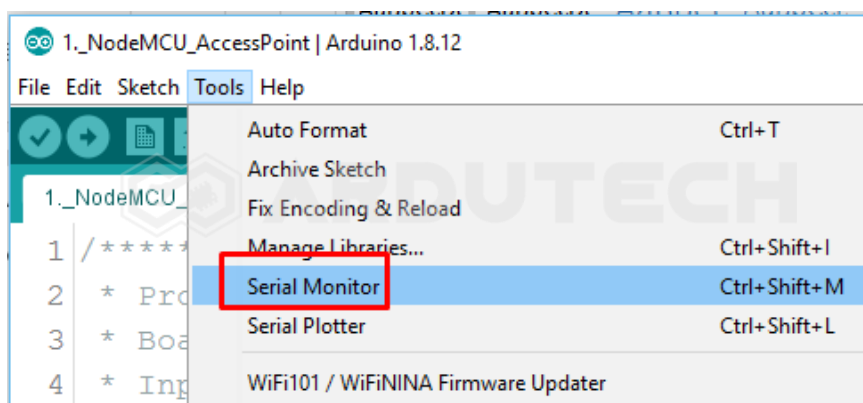
Ganti/sesuaikan variabel berikut :

- Nama jaringan WiFi/hotspot : **WLAN\_SSID**
- Password jaringan WiFi/hotspot : **WLAN\_PASS**
- Username Adafruit IO : **AIO\_USERNAME**
- Active Key Adafruit IO : **AIO\_KEY**

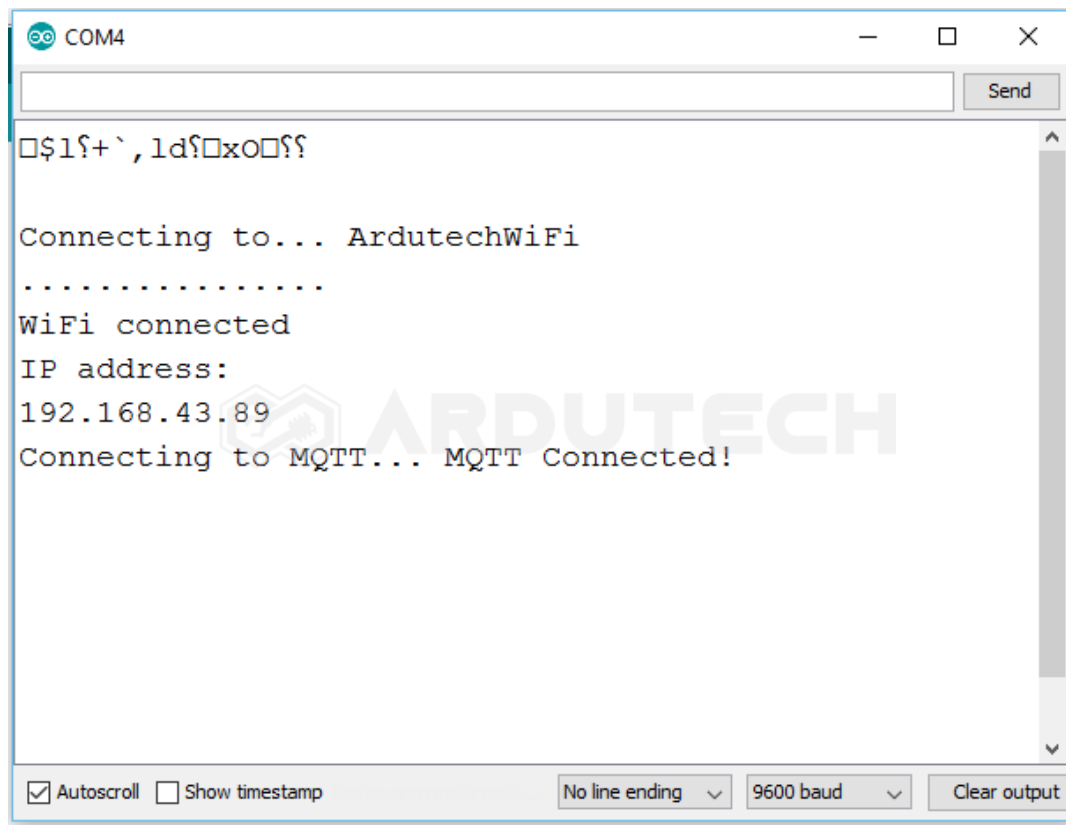
Simpan (**Save**) kemudian **Upload**. Pastikan tidak ada error, jika masih ada silakan cek penulisan dll kemudian perbaiki. (Program ini sudah diuji langsung dan sudah berjalan tanpa ada error)

## Jalannya Alat

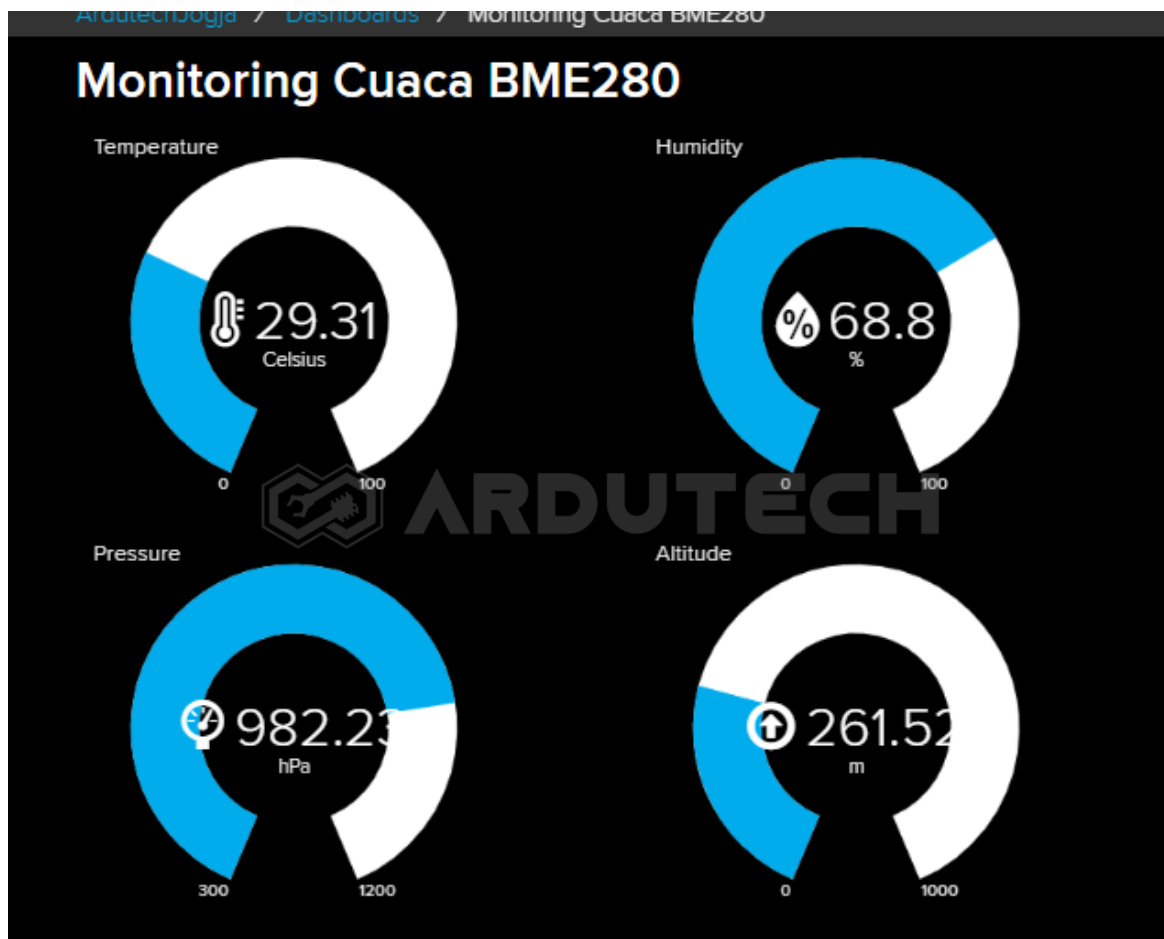
Siapkan jaringan WiFi/hotspot atau tethering dari HP dan aktifkan. Buka Serial Monitor, dari menu **Tools → Serial Monitor** kemudian seting baudrate pada nilai 9600.

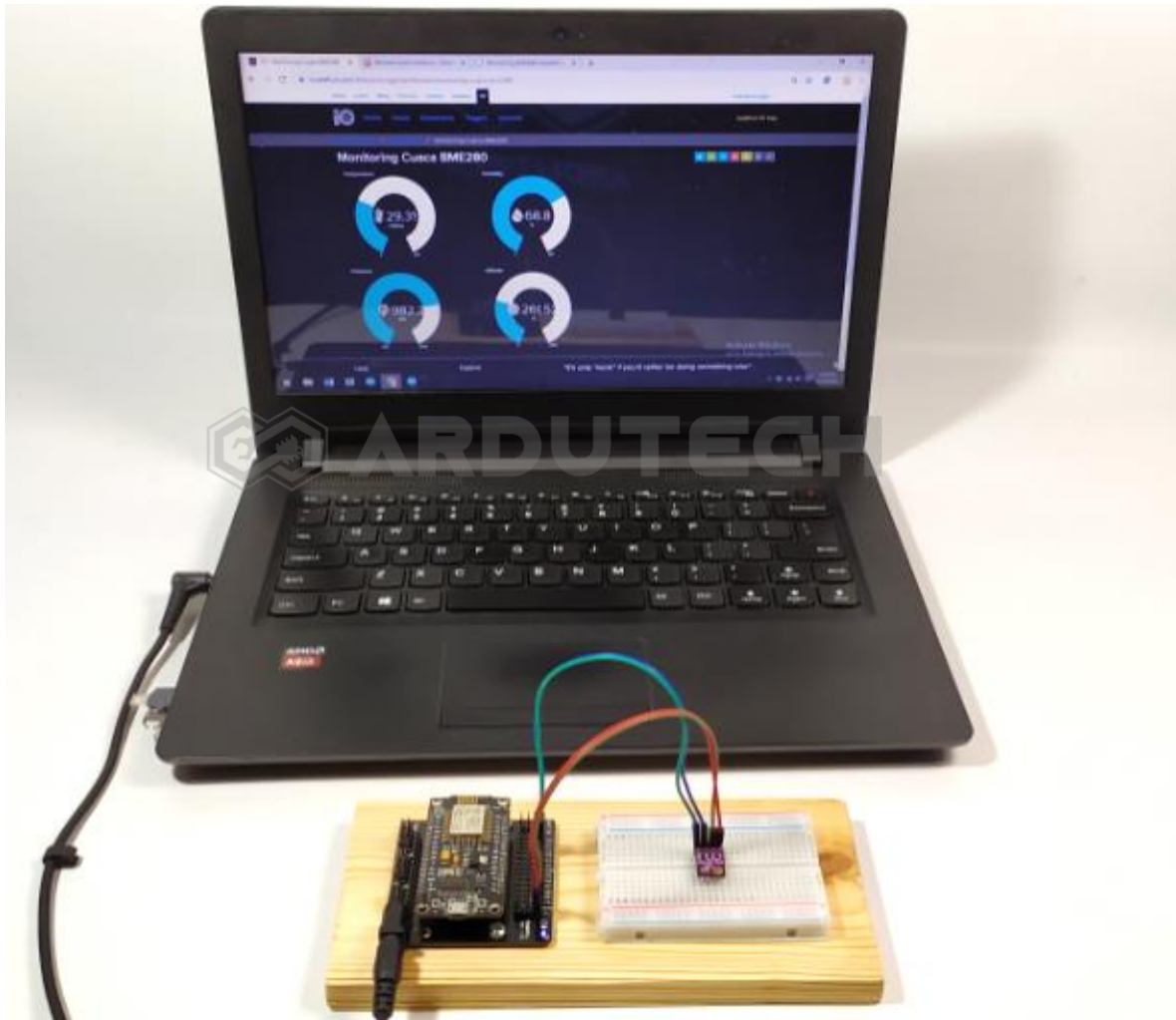


Jika belum muncul tekan tombol reset (RST) pada board NodeMCU sehingga muncul tampilan sebagai berikut :



Setelah berhasil terhubung dengan jaringan WiFi sekarang kita kembali ke tampilan di Adafruit. Perhatikan nilai pada Gauge menunjukkan hasil pembacaan sensor BME280.





Beri perubahan suhu pada sensor BME280 kemudian perhatikan perubahan pada nilai Gauge-nya.

**Selamat berkarya , semoga lancar dan bermanfaat.**

**Ardutech – “Sahabat Inovasi Anda”**